

INDIAN OIL CORPORATION LIMITED // REFINERIES DIVISION

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NOTE FOR APPROVAL (NFA)

File No: IOCL/GHR/HCU-11/NFA/2026-09/042 | Date: 15-September-2026 | Unit: Hydrocracker (HCU-11)  
Initiating Department: Inspection & Asset Integrity Division

SUBJECT: Emergency Repair Sanction — Statutory ASME Thickness Breach on 11-V-102 (1st Stage HP Separator drum)

1. Background & NDT Ultrasonic Survey Defect Discovery

During routine ultrasonic thickness gauging of the 1st Stage High-Pressure Separator (11-V-102), an acute wall thinning anomaly was detected at location **BK-01** (Bottom Knuckle). Base material is **2.25Cr-1Mo + 347 SS cladding**. The field inspector recorded a measured wall thickness of **138.20 mm** against an accelerated corrosion rate of **0.75 mm/yr**.

2. Statutory Code Calculations (ASME Sec VIII Div 1 & API 510)

Verification using the ASME Sec VIII Div 1 UG-27 cylindrical shell formula yielded the following findings:

| Parameter / Statutory Metric           | Engineering Finding & Status             |
|----------------------------------------|------------------------------------------|
| Design Pressure (P)                    | 14.5 MPa (145.0 barg)                    |
| Minimum Required Thickness (t_min)     | 138.57 mm (ASME UG-27)                   |
| Actual Measured Thickness (t_actual)   | 138.20 mm at Point BK-01                 |
| Safety Margin Delta (t_actual - t_min) | -0.37 mm [STATUTORY CODE BREACH]         |
| API 510 Remaining Safe Service Life    | -0.49 Years (BELOW ZERO)                 |
| Recommended Derated MAWP               | 144.6 barg (Immediate operational limit) |

3. CVC Guidelines & Emergency Procurement Justification

In accordance with Central Vigilance Commission (CVC) Circular No. 02/02/2004 and Delegation of Powers (DOP) Clause 4.2 (Single-Source Emergency Procurement under Proprietary Article Certificate), single-source engagement of specialized OEM weld overlay contractor is strictly warranted to prevent catastrophic vessel rupture and loss of secondary containment during continuous sour hydrocracker operation.

4. Recommended Action & Financial Sanction

1. Immediately derate vessel operating pressure from 145.0 barg to 144.6 barg. 2. Execute emergency robotic weld overlay (347 SS) or bottom knuckle replacement during the scheduled October turnaround. 3. Mobilize specialized pre-turnaround NDT radiography and OEM materials immediately.

Estimated Total Expenditure: Rs. 88.0 Lakhs (Covered under Refinery Turnaround Capex Budget Code HCU-2026-CAP).

5. Submission & Approval Block

| Prepared By:                              | Verified By:                                | Approved By:                                                                  |
|-------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------|
| Sr. Inspection Engineer<br>Emp ID: 448102 | Chief Manager (Integrity)<br>Emp ID: 310294 | Executive Director & Refinery Head<br>Competent Financial Authority (DOP 2.1) |